

CCOS Task Catalog

Category	Task Title	One-line Description
First contact & orientation	Route-finding for a new session	If you are new or unsure which module fits your situation, M00 identifies your role and routes you before any clinical data is entered.
First contact & orientation	Compress the case to a problem representation	Before building a differential, compress the case to a one-sentence semantic problem representation. Qualifiers: acuity, severity, time course, host factors.
First contact & orientation	Build the illness script for the leading candidate	Build the formal illness script for the suspected disease and use it as the comparator against your patient's actual findings.
First contact & orientation	Differential diagnosis deep dive	Expand, re-rank, and interrogate the differential. For each entry: what supports it, what excludes it, what investigation would move it up or down.
Diagnostic reasoning	Test each hypothesis deductively	State the leading hypothesis. The AI generates its testable predictions, then checks each against the case data — surfacing where evidence supports, undermines, or is absent.
Diagnostic reasoning	Apply Bayesian likelihood ratios to key findings	For each key finding, estimate pre-test probability and apply likelihood ratios. The AI audits numeric reasoning and shows how each finding shifts the posterior probability.
Diagnostic reasoning	Separate causal from probabilistic reasoning	Run the case through both lenses. The AI surfaces where you are reasoning causally when the data only supports probabilistic inference — and vice versa.
Diagnostic reasoning	Map first-principles pathophysiology	Build the mechanistic chain from underlying biology to clinical presentation. The AI challenges every assumed step and asks for the molecular mechanism.
Diagnostic reasoning	Red herring / signal-to-noise drill	Classify every finding as signal, noise, or red herring. The AI challenges every classification. Use when a case feels complex because of data volume, not genuine ambiguity.
Diagnostic reasoning	Audit your pattern recognition	Name the pattern you fired first. M20 runs the Recognition-Primed Decision model — what cues triggered it, what mental simulation you ran, where System 1 may have short-circuited.
Diagnostic reasoning	Extract and interrogate the diagnostic anchor	Identify the single finding most powerfully anchoring the current diagnosis. Defend it, test its reversibility, and reason through what happens if the anchor is wrong.

Preparation	Ward round preparation — structured SBAR	Prepare the structured SBAR presentation for each patient. The AI acts as a senior peer, probing the problem list, active issues, outstanding investigations, and planned interventions.
Preparation	Overnight trend analysis	Plot the trajectory of key parameters overnight. The AI identifies acceleration, deceleration, plateau, and trend reversal — then asks for clinical interpretation of each velocity change.
Preparation	Real-time case review & data audit	Query and clean the case log before the round. The AI identifies data gaps, inconsistencies, and entries requiring verification — asking you to explain each anomaly.
Preparation	Polypharmacy screen before the round	Run the polypharmacy screen before stepping onto the ward. For each medication: indication, interaction, dose check, and deprescribing opportunity.
On-round decisions	Diagnostic time-out before committing	Pause at the moment of decision. What are we missing? What is the most dangerous diagnosis not yet excluded? Have we anchored prematurely? Assign explicit confidence levels before proceeding.
On-round decisions	High-value care audit of outstanding requests	Audit every outstanding test, investigation, and referral. What is the pre-test probability? What will this change? Does the benefit justify the cost and patient burden?
On-round decisions	"Why now?" — precipitant for acute change	Why did this patient change overnight? The AI will not accept "the disease progressed" — it interrogates for a specific precipitant: physiological, pharmacological, behavioural, or social.
On-round decisions	Discharge planning & throughput strategy	Identify bottlenecks, sequence dependencies, and timing risks in the discharge or ongoing management plan. Build a throughput sequence that maximizes safety without sacrificing efficiency.
On-round decisions	Map certainty across the management plan	For each claim in the plan, assign epistemic status: established fact, evidence-based inference, reasonable assumption, or speculation. Identify where the plan rests on conjecture.
Treatment decisions	Compare treatment options with prognosis trajectories	Compare two or more treatment options: mechanism, evidence quality, response timeline, adverse event profile, and prognostic trajectory. The AI then audits your reasoning for choosing one path.

Treatment decisions	Map the full decision spectrum	For any management decision, map Conservative → Minimal → Balanced → Aggressive → Maximal with explicit rationale and uncertainty flags at each step. Defend your position on the spectrum.
Treatment decisions	Check guideline concordance	Map relevant clinical guidelines, identify where the management plan is concordant and where it deviates, and provide explicit justification for each deviation.
Treatment decisions	Counterfactual stress-test of the plan	Run structured "what if" scenarios. What if the key finding were absent? What if the patient were 20 years older? How does the plan change across each counterfactual?
Safety & harm prevention	Clinical pre-mortem before implementing the plan	Imagine the patient has come to harm six weeks from now. Work backwards to identify the most likely failure modes in the current plan, ranked by probability and severity.
Safety & harm prevention	Iatrogenic domino effect — trace downstream harms	Trace the downstream harm chain for each decision: Treatment A → Adverse Event B → Intervention C → Harm D. Use to stress-test the management plan before committing.
Safety & harm prevention	FMEA — rank failure modes in the management plan	Apply Failure Mode and Effects Analysis to the clinical plan. Enumerate failure modes at each step, estimate severity and likelihood, and prioritize the highest-risk points.
Safety & harm prevention	Health economics & value-based care alignment	Calculate or estimate cost-effectiveness, resource intensity, and value-based care alignment for the proposed plan. The AI challenges assumptions and asks for explicit trade-off reasoning.
Safety & harm prevention	Systems-based clinical analysis	Analyse through a systems lens: patient, team, unit, institutional, and societal factors contributing to this presentation or management risk. Identify intervention points at every level.
Safety & harm prevention	Full cognitive bias audit	Systematic audit for anchoring, availability, premature closure, framing effects, and commission bias. The AI names each bias, its mechanism, and the specific moment it appears in the reasoning.
Evidence integration	Map the evidence frontier for a clinical question	What is known, what is contested, what is recent, and where the evidence base is thin or absent — for this diagnosis or management question.
Evidence integration	Evidence-based medicine appraisal	Apply the Sackett EBM cycle: formulate the PICO, identify the best evidence, critically appraise, integrate. Then run the adversarial counterpart to challenge the appraisal under pressure.

Acute deterioration workup	Find the precipitant — "Why now?"	The AI will not accept "the disease progressed." It interrogates for a specific precipitant — physiological, pharmacological, behavioural, social — and audits whether you have actually looked for it.
Acute deterioration workup	Poly-crisis — map how pathologies are interacting	For patients where multiple pathologies are simultaneously active, M38 simulates how each problem interacts with and amplifies the others — identifying the single intervention most likely to interrupt the cascade.
Acute deterioration workup	Velocity analysis — what is deteriorating fastest?	Plot the trajectory of key clinical parameters. The AI identifies which parameters are accelerating, which are plateauing, and asks you to interpret the clinical significance of each rate of change.
Acute deterioration workup	Is treatment contributing to the deterioration?	The Domino Effect traces how prior treatment decisions may have contributed to the current state. The Medication Reconciliation finds pharmacological risks already embedded in the chart.
Acute deterioration workup	Anticipate the next failure point	FMEA identifies where the next failure is most likely. The Pre-Mortem imagines it has occurred and asks you to trace the mechanism — so you can interrupt it before it happens.
Acute deterioration workup	Emergency diagnostic time-out	Before committing to a management plan for a deteriorating patient, pause and run the Time-Out + Bias Audit. What dangerous diagnosis have we not yet excluded?
Acute deterioration workup	What if the working diagnosis is wrong?	Extract the anchor, then run counterfactuals designed specifically to break it. What would the case look like if the anchor finding were absent, misleading, or a red herring?
Acute deterioration workup	Triage and sequence competing priorities	For a patient with multiple simultaneous clinical priorities, build a throughput sequence that addresses the highest-risk first without creating new bottlenecks elsewhere.
Handover preparation	Overnight triage anticipation — prepare the covering team	Rehearse the night-cover scenario: which patient is seen first, which can wait, which needs immediate escalation, and which needs an anticipatory care plan before you leave.
Handover preparation	Outstanding risks and uncertainty flags for handover	Map epistemic certainty across the management plan and run a final Time-Out before handover. What is uncertain? What would trigger escalation? What must the covering team know?
Handover preparation	Set escalation triggers and implementation plan	Apply implementation science to the ongoing management plan: identify barriers, enablers, feedback loops, and explicit escalation criteria for the covering team.

Case closure & learning	Distill clinical pearls from a resolved case	After a case resolves, distill the key learning points into shareable clinical pearls — then stress-test each pearl for generalizability and exception conditions.
Case closure & learning	Post-adverse-event structured review	When something has gone wrong, run a structured post-event analysis: FMEA to rank failure modes, the Domino Effect to trace the harm mechanism, and Systems-Based Analysis for systemic factors.
Case closure & learning	Produce the diagnostic reasoning map	Generate a structured map of the complete diagnostic reasoning process: anchor findings, branch points, pivots, and the final diagnosis. Documents how the reasoning evolved, not just where it landed.
Case closure & learning	Thematic analysis of the case narrative	Identify the recurring clinical themes across the case history. The AI names the dominant themes, asks whether they form a coherent narrative, and surfaces data points that do not fit.
Diagnostic pipelines	Complete Diagnostic Pipeline	From presentation to diagnosis with full cognitive audit. For complex, uncertain, or high-stakes cases.
Diagnostic pipelines	Cognitive Safety & Bias Pipeline	Complete audit of how you are thinking, not just what. For cases where you have a strong intuitive read and want to audit it before acting.
Management & safety pipelines	Management Decision Pipeline	From confirmed diagnosis to defensible management plan. For complex decisions, polypharmacy, or any case with significant iatrogenic potential.
Management & safety pipelines	Patient Safety Pipeline	Systematic pre-action harm reduction for high-risk interventions, polypharmacy, or complex multi-morbid patients. "First, do no harm" as a structured workflow.
Management & safety pipelines	Deteriorating Patient / Complex Systems Pipeline	For patients in whom everything is going wrong simultaneously: multi-organ failure, frail elderly with multiple active problems, ICU complexity.
Evidence & ward round pipelines	Evidence-Informed Practice Pipeline	From clinical question to evidence-integrated decision. For cases where you plan to deviate from standard management, or need to integrate recent literature.
Evidence & ward round pipelines	Complete Ward Round Pipeline	From overnight events to post-round plan. Can be run for a single patient or across a patient list.
Evidence & ward round pipelines	Uncertainty & Epistemic Calibration Pipeline	When you know what you don't know and need to map it rigorously. For diagnostically ambiguous presentations, or when the team disagrees.

Advocate workflows	Build a structured case record as an advocate	Four-step workflow for a family member to build a structured case record: timeline, medication log, question list, and consultation summary.
Advocate workflows	Longitudinal monitoring and tracking	Extends M02 into ongoing tracking: daily symptoms, mood, medication adherence, lifestyle factors, and red-flag alerts. The AI organizes observations and identifies trends.
Advocate workflows	Patient needs assessment	Run a structured assessment of what the patient actually needs — biomedical, psychological, social, and informational. Distinguish between what is being treated and what the patient experiences as their most pressing problem.
Advocate workflows	Answer patient questions clearly and accurately	For a patient or advocate question, M11 constructs a health-literacy-appropriate response then audits it for accuracy, completeness, and the risk of misinterpretation.
Advocate workflows	Identify community supports and social resources	Analyse the social determinants in the case and identify community-level supports, services, and resources the patient can access. Use to plan post-discharge support.
Advocate workflows	Complete Patient Advocate Pipeline	Full support workflow for a family member or non-clinician advocate: documentation → monitoring → needs assessment → education → medication review → community planning.
Self-directed learning	Socratic case interrogation — reason before receiving	Present a case. The AI withholds its assessment and forces you to commit to and justify your reasoning before any answer is given. Works at any level of seniority.
Self-directed learning	Acquire an illness script for a new disease	Build the formal illness script, then run M16 to integrate the underlying basic science — so the script is grounded in pathophysiology, not just a checklist.
Self-directed learning	Critical appraisal of a journal article	Walk through structured critical appraisal: study design, bias risk, primary endpoint, effect size, confidence intervals, and applicability to a specific patient or question.
Self-directed learning	Apply global & genetics knowledge to a complex case	For unusual or diagnostically challenging cases, run the Global Knowledge Network Matrix and Clinical Genetics module to surface international evidence and hereditary considerations.
Trainee assessment & feedback	Structured self-assessment with Bloom's audit	Run M01 for case reasoning, then apply Bloom's Taxonomy to identify which cognitive levels were reached and FLINK to ensure learning was meaningful — not just correct.

Trainee assessment & feedback	Prepare a case for teaching	Compress the case, generate a graded question set for the session, and distill the clinical pearls — producing structured teaching material with both System 1 and System 2 questions.
Trainee assessment & feedback	N-of-1 research protocol for an unusual case	Run the seven-stage N-of-1 research protocol: case definition, hypothesis, data collection plan, intervention logic, outcome measurement, analysis, and dissemination readiness.
Trainee assessment & feedback	Resource-constrained reasoning drill	Re-run diagnostic and management reasoning under defined resource constraints. Tests whether your reasoning holds when the standard algorithm is stripped of investigations and specialist access.
Trainee assessment & feedback	Complete Clinical Cognition Science Pipeline	A full deep-dive into how expert clinicians think. Covers the complete spectrum from intuitive pattern recognition to formal epistemic mapping.